

11-mavzu. n-tartibli chiziqli differensial tenglamalar. Vronskiy determinanti. Fundamental yechimlar sistemasi. (2-soat).

Reja:

1. n-tartibli chiziqli differensial tenglamalar va ularning umumiy xossalari.
2. Umumiy yechimning xossalari.
3. Mavjudlik va yagonalik teoremasi.

Chiziqli erkli funksiyalar, Chiziqli bog'langan funksiyalar, *chiziqli erkli*, *chiziqli bog'langan*, *Vronskiy determinanti differensial operator*, *chiziqli kombinatsiya*

Biz bu yerda ushbu

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = g(x) \quad (1)$$

n - tartibli chiziqli differensial tenglama yechimlarining umumiy xossalarini o'rganamiz; bunda $y = y(x)$ - noma'lum funksiya, berilgan $a_{n-1}(x), \dots, a_1(x), a_0(x)$ koefitsientlar va $g(x)$ ozod had (o'ng tomon) biror I oraliqda aniqlangan va uzluksiz deb hisoblanadi, ya'ni $\{a_{n-1}(x), \dots, a_1(x), a_0(x), g(x)\} \subset C(I)$.

Quyidagi Koshi shartlarini (boshlang'ich shartlarni) qaraylik:

$$y|_{x_0} = y_0, y'|_{x_0} = y'_0, \dots, y^{(n-1)}|_{x_0} = y_0^{(n-1)}; \quad (2)$$

bunda $x_0 \in I$ va $y_0, y'_0, \dots, y_0^{(n-1)}$ berilgan ixtiyoriy sonlar.

Yuqorida e'tirof etilganidek, aytilgan shartlarda (II.4.1),(II.4.2) Koshi masalasi birato'la I oraliqda aniqlangan yechimga ega va bunday yechim yagonadir.

(II.4.1) tenglamaning ozod hadi o'rniga 0 qo'yib, (II.4.1) ga mos bir jinsli tenglamani hosil qilamiz:

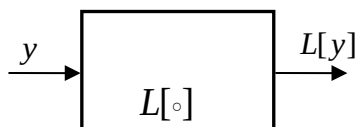
$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = 0. \quad (2)$$

Ravshanki, bu bir jinsli tenglama $y = 0$ trivial yechimga ega.

Qulaylik uchun ushbu

$$L[y] \stackrel{\text{def}}{=} y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y \quad (4)$$

n - tartibli differensial operatorni kiritaylik. Biz bu operatorni I oraliqda n marta uzluksiz differensiallanuvchi funksiyalarga ta'sir ettiramiz, ya'ni (II.4.3) operatorni $y = y(x) \in C^n(I)$ funksiyalarda aniqlangan deb hisoblaymiz: $L[0]: C^n(I) \rightarrow C(I)$. $L[\circ]$ operator kirish funksiyasi y ga ko'ra chiqish funksiyasi $L[y]$ ni hisoblaydi (aniqlaydi):



Jumla. (4) differensial operator chiziqlidir, ya'ni

$$a) \forall \{y_1, y_2\} \subset C^n(I) \quad L[y_1 + y_2] = L[y_1] + L[y_2] \quad (\text{additivlik})$$

$$b) \forall \lambda \in \mathbb{R} \quad \forall y \in C^n(I) \quad L[\lambda \cdot y] = \lambda \cdot L[y] \quad (\text{bir jinslilik})$$

xossalari o'rinli.

□ Isboti hosilaning chiziqlilik xossasidan bevosita kelib chiqadi. □

Natija. Agar, $\{y_1, y_2, \dots, y_k\} \subset C^n(I)$ va $\{c_1, \dots, c_k\} \subset \mathbb{R}$ bo'lsa, u holda

$$L \left[\sum_{j=1}^k c_j y_j \right] = \sum_{j=1}^k c_j L[y_j]$$

tenglik o'rinli.

$y_1, y_2, \dots, y_k$ **funksiyalarning chiziqli kombinatsiyasi** deb ushbu $c_1 y_1 + c_2 y_2 + \dots + c_k y_k$ yig'indiga aytiladi; bunda $\{c_1, \dots, c_k\} \subset \mathbb{R}$.

Teorema. (II.4.1₀) chiziqli bir jinsli differensial tenglama yechimlarining chiziqli kombinatsiyasi yana shu tenglamaning yechimidir.

□ $y_1, y_2, \dots, y_k$ (II.4.1₀) ning yechimlari bo'lsin: $L[y_j] = 0, j = \overline{1, k}$. Biz ularning

$\sum_{j=1}^k c_j y_j$ chiziqli kombinatsiya ham (II.4.1₀) tenglamani qanoatlantirishini tekshirishimiz kerak. Bu esa natija va berilganga ko'ra ravshan:

$$L \left[\sum_{j=1}^k c_j \cdot y_j \right] = \sum_{j=1}^k c_j \cdot L[y_j] = \sum_{j=1}^k c_j \cdot 0 = 0. \quad \square$$

(II.4.1₀) tenglamaning barcha yechimlari to'plamini V_n bilan belgilaylik:

$$V_n = \{y \in C^n(I) \mid L[y] = 0\}. \quad (\text{II.4.4})$$

V_n da funksiyalarni qo'shish va funksiyani songa ko'paytirish amallarini odatdagicha (nuqtama-nuqta) kiritamiz.

Mavzuni mustahkamlash uchun savollar:

1. n-tartibli chiziqli differensial tenglamalarning qanday xossalar bor?
2. Umumiy yechimning qanday xossalari bor?
3. Mavjudlik va yagonalik teoremasi qanday isbotlanadi?